Mango-StorageBoost-2x-NTI-NTT-221H-221H-1-client

Version of benchmark: MLPerf Storage v2.0

Solution type: Fabric-attached block device

Submission devision: CLOSED

MangoBoost

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1 Hardware

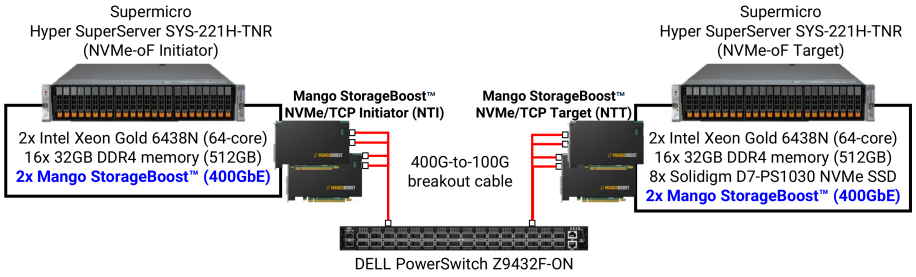


Figure 1: System under test.

Figure 1 shows the system under test. One client and one storage server are connected to the same switch. The client and server has 400G aggregate bandwidth with four 100G ports. On both, client and server sides, MangoBoost StorageBoost™ NVMe/TCP Initiator (NTI) and MangoBoost StorageBoost™ NVMe/TCP Target (NTT) are deployed separately to offload NVMe/TCP processing from the host CPU to the DPU.

1.1 Client

The client is a commercial Supermicro Hyper SuperServer SYS-221H-TNR. The MangoBoost DPUs in the server provides NVMe/TCP initiator offloading.

1.2 Server

The server is a commercial Supermicro Hyper SuperServer SYS-221H-TNR with 8x Solidigm D7-PS1030 NVMe SSDs. The MangoBoost DPUs in the server provides NVMe/TCP target offloading.

1.3 Switch

The switch is a commercial DELL PowerSwitch Z9432F-ON.

1.4 Power Requirements

System component	Rack Units	PSU	Nameplate rated power	Design Power
Storage Server	2	PS 1	1200 watts	1200 watts
		PS 2	1200 watts	
Totals	**2 units**		**2400 watts**	**1200 watts**

2 Software

2.1 Client

The OS is Ubuntu 22.04.4 LTS (GNU/Linux 5.15.0-143-generic x86_64). The NVMe/TCP connections are made by Mango StorageBoost™ NTI and exposed as local NVMe devices to the host. Mango StorageBoost™ NTI is fully compatible with the inbox Linux NVMe driver.

2.2 Server

The OS is Ubuntu 22.04.4 LTS (GNU/Linux 5.15.0-143-generic x86_64). The server uses modified Storage Performance Development Kit v23.9 (SPDK) to provide Mango StorageBoost NVMe/TCP target offloading driver and NVMe/TCP target service to the client.

3 Settings

3.1 Client

NTI makes NVMe/TCP connections in the on-board ARM processor. The connected NVMe devices are exposed as local NVMe devices to the host at `/dev/nvme*`.

We configured RAID0 to fully exploit the aggregate performance of NVMe SSDs.

```
sudo pvcreate /dev/nvme*n1
sudo vgcreate lvm_raid /dev/nvme*n1
sudo lvcreate --extents 100%FREE --name lvm_raid lvm_raid -i16 -I64k

sudo mkfs.ext4 /dev/lvm_raid/lvm_raid
sudo mount /dev/lvm_raid/lvm_raid /mnt
```

3.2 Server

The server exposes NVMe devices using MangoBoost SPDK software. MangoBoost SPDK binds NVMe devices to DPU and then exposes them to clients through NVMe/TCP. In the SUT, 2 NVMe devices are assigned to each of the four ports in the server.